

Module 2(b) : Discrete Fourier Transforms (DFT)

DFT : Frequency domain sampling and reconstruction of discrete time signals. DFT as a linear transformation, its relationship with other transforms. Properties of DFT, Multiplication of two DFTs- the Circular convolution.

Introduction: Digital Signal Processing (DSP) refers to processing of signals by digital systems like Personal Computers (PC) and systems designed using digital Integrated Circuits (ICs), Microprocessors, microcontrollers etc. In DSP, all the signals are discrete and the analog signals such as speech, music, audio etc are converted to discrete form by Analog to Digital Converters (ADC) and after processing, the signals are converted back to analog form using Digital to Analog Converters (DAC).

Many applications demand the processing of signals in frequency domain. For example, frequency content, energy, power spectrums etc, can be better analyzed in frequency domain. Hence signals are transformed from time domain to frequency domain using Fourier Transforms (FT) and Discrete Fourier Transforms (DFT). For discrete signals DFT is used. DFT can also be used for linear filtering and correlation. Once the required analysis and processing is done, then the signals are transformed back to time domain by Inverse Discrete Fourier Transform (IDFT).

Frequency Domain Sampling and Reconstruction of Discrete time signals

Need of Frequency Domain Sampling

The Fourier transform $X(\Omega)$ of a discrete time signal $x(n)$ is given as

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\Omega n} \quad \text{--- (1)}$$

where Ω is the discrete frequency and Ω is the continuous function from 0 to 2π .

Even though $x(n)$ is a discrete time signal, its spectrum is periodic and continuous in nature, hence it cannot be evaluated using digital computer / digital processor.

Frequency Domain Sampling

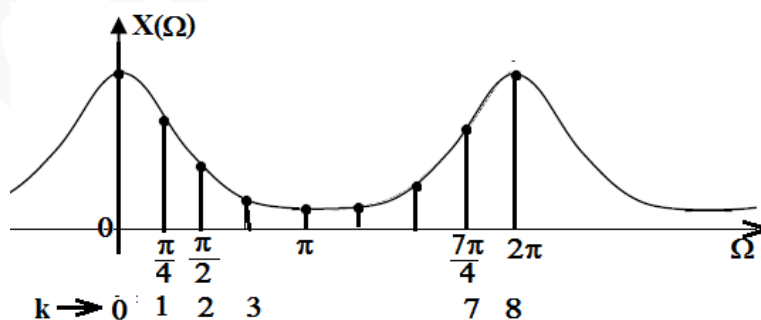
To overcome the above problem, the spectrum of $X(\Omega)$ is sampled uniformly. Let N samples are taken from 0 to 2π and hence the spacing between the samples $\left(\frac{2\pi}{N}\right)$.

Using $\Omega = \left(\frac{2\pi}{N}k\right)$ in equation (1) results

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\left(\frac{2\pi}{N}k\right)n} \quad \text{--- (2)}$$

Where k is the index for samples and $k = 0, 1, 2, 3, \dots, (N-1)$

Figure shows the samples of $X(\Omega)$ for $N = 8$.



$k = 0; \Omega = \left(\frac{2\pi}{N}\right)k = \frac{2\pi}{8}(0) = 0$	$k = 2; \Omega = \frac{\pi}{2}$	$k = 4; \Omega = \pi$	$k = 6; \Omega = \frac{3\pi}{2}$
$k = 1; \Omega = \left(\frac{2\pi}{N}\right)k = \frac{2\pi}{8}(1) = \frac{\pi}{4}$	$k = 3; \Omega = \frac{3\pi}{4}$	$k = 5; \Omega = \frac{5\pi}{4}$	$k = 7; \Omega = \frac{7\pi}{4}$

To determine Minimum value of N

In equation (2), n varies from $(-\infty \text{ to } +\infty)$. Divide the summation into individual summations containing only N samples of x(n).

$$X\left(\frac{2\pi}{N} k\right) = \dots + \sum_{n=-N}^{-1} x(n)e^{-j\left(\frac{2\pi}{N}\right)kn} + \sum_{n=0}^{N-1} x(n)e^{-j\left(\frac{2\pi}{N}\right)kn} + \sum_{n=N}^{2N-1} x(n)e^{-j\left(\frac{2\pi}{N}\right)kn} + \dots$$

The above summation can be represented as

$$X\left(\frac{2\pi}{N} k\right) = \sum_{l=-\infty}^{\infty} \sum_{n=lN}^{lN+N-1} x(n)e^{-j\left(\frac{2\pi}{N}\right)kn}$$

If the index of the inner summation is changed from n to $(n - lN)$; then the limits will be $n = 0$ and $n = (N - 1)$

$$X\left(\frac{2\pi}{N} k\right) = \sum_{l=-\infty}^{\infty} \sum_{n=0}^{N-1} x(n - lN)e^{-j\left(\frac{2\pi}{N}\right)kn}$$

$$\begin{aligned} \text{when } l = 0; & \sum_{n=0}^{N-1} x(n)e^{-j\left(\frac{2\pi}{N}\right)kn} \\ \text{when } l = -1; & \sum_{n=-N}^{-1} x(n)e^{-j\left(\frac{2\pi}{N}\right)kn} \\ \text{when } l = 1; & \sum_{n=N}^{2N-1} x(n)e^{-j\left(\frac{2\pi}{N}\right)kn} \end{aligned}$$

Interchanging the order of summation results

$$X\left(\frac{2\pi}{N} k\right) = \sum_{n=0}^{N-1} \sum_{l=-\infty}^{\infty} x(n - lN)e^{-j\left(\frac{2\pi}{N}\right)kn} = \sum_{n=0}^{N-1} x_p(n) e^{-j\left(\frac{2\pi}{N}\right)kn} \quad \text{--- (A)}$$

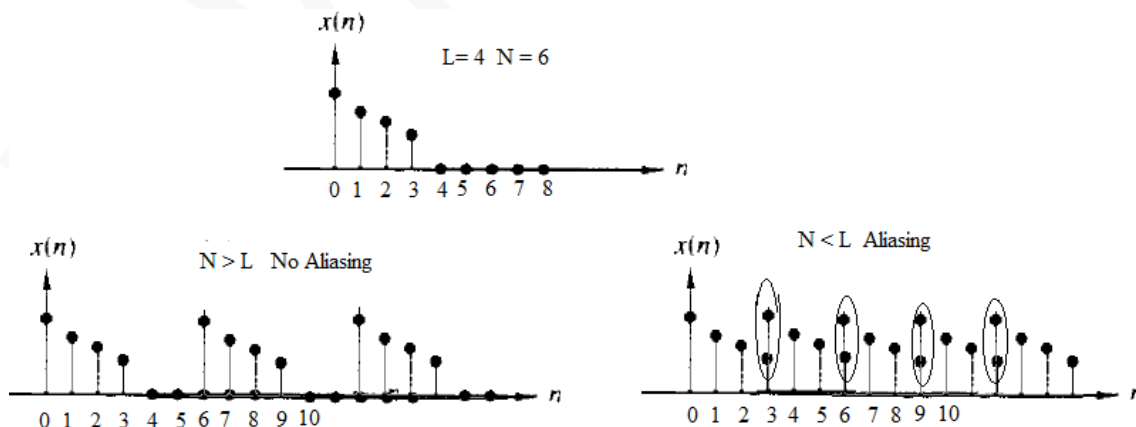
where $x_p(n) = \sum_{l=-\infty}^{\infty} x(n - lN) = \dots + x(n + 2N) + x(n + N) + x(n) + x(n - N) + x(n - 2N) + \dots$

$x_p(n)$ is the periodic repetition of $x(n)$ with a the period of N samples.

Let $x(n)$ be a non periodic signal with L samples. Signal $x_p(n)$ is periodic with period N

Case 1: $N > L$: No aliasing :

Let $L = 4$ and $N = 6$; Observe that $x(4)$ and $x(5)$ are zeros. This is because the signal repeats at 6, 12.... Etc. Since $N > L$, no aliasing (overlapping of signal)



Case2: $N < L$: Aliasing :

Let $N = 3$, since $N < L$, there are two samples are overlapping at $n = 3, 6, \dots$. This called aliasing and it is not possible to obtain $x(n)$ from $x_p(n)$.

To avoid aliasing in time domain, the number of samples in the frequency spectrum must be greater than the number of samples in time domain sequence, i.e $N > L$

Reconstruction:

The periodic signal $x_p(n)$ can be expressed by discrete Fourier series as

$$x_p(n) = \sum_{k=0}^{N-1} c_k e^{j\left(\frac{2\pi}{N}\right)kn} ; \quad n = 0,1,2, \dots (N-1) \text{ --- (B)}$$

And the Fourier coefficient $c_k = \frac{1}{N} \sum_{n=0}^{N-1} x_p(n) e^{-j\left(\frac{2\pi}{N}\right)kn} ; \quad k = 0,1,2 \dots (N-1) \text{ --- (C)}$

From equations (A) and (C)

$$c_k = \frac{1}{N} X\left(\frac{2\pi}{N} k\right) = \frac{X[k]}{N} \quad \text{where } k = 0,1,2 \dots (N-1)$$

Hence equation (C) can be written as

$$x_p(n) = \sum_{k=0}^{N-1} \frac{1}{N} X\left(\frac{2\pi}{N} k\right) e^{j\left(\frac{2\pi}{N}\right)kn} \quad n = 0,1,2, \dots (N-1) \text{ --- (D)}$$

Equation (D) gives the expression for time domain sequence $x_p(n)$ from frequency domain samples $X\left(\frac{2\pi}{N} k\right)$.
Hence the time domain sequence can be constructed from frequency domain sampling.

1.2(a) - Definition of DFT

Let $x(n)$ be a discrete signal of length L , $X(k)$ be the DFT of $x(n)$

The N point DFT of $x(n)$ where $N \geq L$ is defined as

$$X\left(\frac{2\pi}{N} k\right) = X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi}{N}\right)kn} \quad \text{for } k = 0,1,2,3 \dots \dots (N-1) \text{ --- (1)}$$

Equation (1) transforms the sequence $x(n)$ of length $L \leq N$ into a sequence of frequency samples $X(k)$ of length N . Since the frequency samples are obtained by evaluating the FT $X(\Omega)$ at set of N (equally spaced) discrete frequencies, the relation is called Discrete Fourier Transform (DFT) of $x(n)$.

1.2(b) - Definition of IDFT

Let $x(n)$ be a discrete signal of length L , $X(k)$ be the N -point DFT of $x(n)$

The Inverse DFT of sequence $X(k)$ of length N is defined as

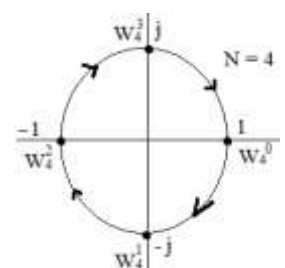
$$\text{IDFT}[X(k)] = x(n) = \sum_{k=0}^{N-1} \frac{1}{N} X(k) e^{j\left(\frac{2\pi}{N}\right)kn} \quad n = 0,1,2, \dots (N-1) \text{ --- (2)}$$

Equation (2) allows us to recover the sequence $x(n)$ from frequency samples and is called Inverse Discrete Fourier Transform (IDFT).

1.3 – Twiddle Factor W_N

$$\text{Let } W_N = e^{-j2\pi/N} = \cos\left(\frac{2\pi}{N}\right) - j\sin\left(\frac{2\pi}{N}\right)$$

W_N is a complex quantity and is called Twiddle factor and hence DFT and IDFT equation become



$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} \quad k = 0, 1, 2, 3, \dots, (N-1)$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} \quad n = 0, 1, 2, 3, \dots, (N-1)$$

W_N can be expressed as $|W_N| = 1, \angle(2\pi/N)$

1.3 (a) - Properties of Twiddle Factor

(1) Periodicity : $W_N^{k+N} = W_N^k$

Proof: $W_N^{k+N} = e^{-j(2\pi/N)(k+N)}$

$$W_N^{k+N} = e^{-j(2\pi k/N)} \cdot e^{-j(2\pi N/N)} = W_N^k (1) = W_N^k$$

(2) Symmetry property: $W_N^{k+N/2} = -W_N^k$

Proof: $W_N^{k+N/2} = W_N^k W_N^{N/2} = W_N^k e^{-j(2\pi k/2N)}$

$$W_N^{k+N/2} = W_N^k e^{-j\pi k} = W_N^k (-1)^k = -W_N^k$$

(3) Conjugation: $W_N^* = W_N^{-1}$

Proof: $W_N^* = [e^{-j(2\pi/N)}]^* = [e^{j(2\pi/N)}] = W_N^{-1}$

(4) Modifying the period : $W_N^{2k} = W_{N/2}^k$

Proof: $W_N^{2k} = e^{-j2(2\pi k/N)} = e^{-j(2\pi k/(N/2))} = W_{N/2}^k$

Note: Twiddle factor values for $N = 4$

$$[W_N]^{kn} = [W_4]^{kn}$$

$$(kn) = 0 \Rightarrow [W_4]^0 = W_4^0 = 1$$

$$\text{For } (kn) = 1 \Rightarrow W_4^1 = e^{-j2\pi/4} = e^{-j\pi/2} = -j$$

$$\text{For } (kn) = 2 \Rightarrow W_4^2 = e^{-j4\pi/4} = e^{-j\pi} = -1$$

$$\text{For } (kn) = 3 \Rightarrow W_4^3 = e^{-j6\pi/4} = e^{-j3\pi/2} = j$$

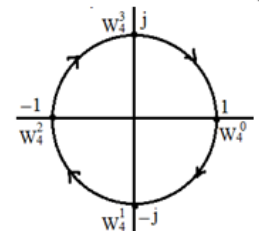
Note:

$$W_4^4 = W_4^{(0+4)} = W_4^0 \quad W_4^4 = W_4^0$$

$$W_4^5 = W_4^{(1+4)} = W_4^1 \quad W_4^5 = W_4^1$$

$$W_4^6 = W_4^{(2+4)} = W_4^2 \quad W_4^6 = W_4^2$$

$$W_4^7 = W_4^{(3+4)} = W_4^3 \quad W_4^7 = W_4^3$$



1.3(b) - DFT as a Linear Transformation

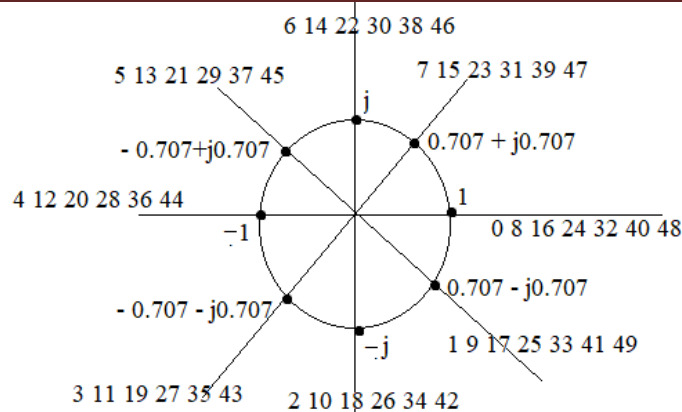
The DFT $X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn}$ $k = 0, 1, 2, 3, \dots, (N-1)$ can be expressed as

$$X[k] = \begin{bmatrix} W_N^0 & W_N^0 & W_N^0 & \dots & W_N^0 \\ W_N^0 & W_N^1 & W_N^2 & \dots & W_N^{N-1} \\ W_N^0 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ W_N^0 & W_N^{N-1} & W_N^{2(N-1)} & \dots & W_N^{(N-1)(N-1)} \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ \vdots \\ x(N-1) \end{bmatrix}$$

For $N = 4$, the Twiddle factor Matrix is

$$W_4^{kn} = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

	0	1	2	3	4	5	6	7
0	W_8^0	W_8^0	W_8^0	W_8^0	W_8^0	W_8^0	W_8^0	W_8^0
1	W_8^0	W_8^1	W_8^2	W_8^3	W_8^4	W_8^5	W_8^6	W_8^7
2	W_8^0	W_8^2	W_8^4	W_8^6	W_8^8	W_8^{10}	W_8^{12}	W_8^{14}
3	W_8^0	W_8^3	W_8^6	W_8^9	W_8^{12}	W_8^{15}	W_8^{18}	W_8^{21}
4	W_8^0	W_8^4	W_8^8	W_8^{12}	W_8^{16}	W_8^{20}	W_8^{24}	W_8^{28}
5	W_8^0	W_8^5	W_8^{10}	W_8^{15}	W_8^{20}	W_8^{25}	W_8^{30}	W_8^{35}
6	W_8^0	W_8^6	W_8^{12}	W_8^{18}	W_8^{24}	W_8^{30}	W_8^{36}	W_8^{42}
7	W_8^0	W_8^7	W_8^{14}	W_8^{21}	W_8^{28}	W_8^{35}	W_8^{42}	W_8^{49}



Twiddle factor Matrix for DFT with N = 8

1	1	1	1	1	1	1	1
1	$\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-j	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	j	$\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$
1	-j	-1	j	1	-j	-1	j
1	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	j	$\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-1	$\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	-j	$-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$
1	-1	1	-1	1	-1	1	-1
1	$-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	-j	$\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	-1	$\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	j	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$
1	j	-1	-j	1	j	-1	-j
1	$\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	j	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-j	$\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$

Twiddle factor Matrix for DFT with N = 4

1	1	1	1
1	-j	-1	j
1	-1	1	-1
1	j	-1	-j

Twiddle factor Matrix for IDFT with N = 8

1	1	1	1	1	1	1	1
1	$\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	j	$-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-j	$\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$
1	j	-1	-j	1	j	-1	-j
1	$-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	-j	$\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	-1	$\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	j	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$
1	-1	1	-1	1	-1	1	-1
1	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	j	$\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-1	$\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	-j	$-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$
1	-j	-1	j	1	-j	-1	j
1	$\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-j	$-\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}}$	-1	$-\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$	j	$\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}}$

Twiddle factor Matrix for IDFT with N = 4

1	1	1	1
1	j	-1	-j
1	-1	1	-1
1	-j	-1	j

Numerical Problems:

1. Show that $\sum_{n=0}^{N-1} W_N^{kn} = N\delta(k) = \begin{cases} N; & k = 0 \\ 0; & k \neq 0 \end{cases}$

Solution: $\sum_{n=0}^{N-1} W_N^{kn} = \sum_{n=0}^{N-1} [W_N^k]^n = \frac{W_N^{kN} - 1}{W_N^k - 1} = \frac{e^{-j2\pi k} - 1}{e^{-j2\pi k/N} - 1}$

Since k is an integer, for $k \neq 0$, the numerator value is 0 and hence the summation becomes zero. for $k = 0$;

$$\sum_{n=0}^{N-1} W_N^{kn} = \sum_{n=0}^{N-1} W_N^0 = \sum_{n=0}^{N-1} 1 = N|_{k=0} = N\delta(k)$$

Similarly, $\sum_{n=0}^{N-1} W_N^{(k-k_0)n} = N|_{k=k_0} = N\delta(k-k_0)$ and $\sum_{n=0}^{N-1} W_N^{(k+k_0)n} = N|_{k=-k_0} = N\delta(k+k_0)$

2. Compute the N – point DFT of the following signals: (i) $x(n) = \delta(n)$ (ii) $x(n) = \delta(n - n_0)$
(iii) $x(n) = a^n$ and compute the DFT value of $x(n) = 0.5^n$; $0 \leq n \leq 3$ (iv) $x(n) = e^{j\frac{2\pi k_0 n}{N}}$

(v) $x(n) = \begin{cases} 1; & 0 \leq n \leq (\frac{N}{2} - 1) \\ 0; & \frac{N}{2} \leq n \leq (N - 1) \end{cases}$ (N even) (vi) $x(n) = \cos(\frac{2\pi k_0 n}{N})$ (vii) $x(n) = \sin(\frac{2\pi k_0 n}{N})$

(i) $x(n) = \delta(n)$

Solution: $x(n) = \delta(n) = \begin{cases} 1; & n = 0 \\ 0; & n \neq 0 \end{cases}$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j(\frac{2\pi}{N})kn} = \sum_{n=0}^{N-1} \delta(n) e^{-j(\frac{2\pi}{N})kn} = 1 \text{ at } n = 0$$

Hence $\delta(n) \xleftarrow{\text{DFT}_N} 1$

(ii) $x(n) = \delta(n - n_0)$

Solution: $x(n) = \delta(n - n_0) = \begin{cases} 1; & n = n_0 \\ 0; & n \neq n_0 \end{cases}$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j(\frac{2\pi}{N})kn} = \sum_{n=0}^{N-1} \delta(n - n_0) e^{-j(\frac{2\pi}{N})kn} = \left[e^{-j(\frac{2\pi}{N})kn_0} \right]_{n=n_0} = W_N^{kn_0}$$

Similarly $\delta(n + n_0) \xleftarrow{\text{DFT}_N} W_N^{-kn_0}$

(iii) $x(n) = a^n$ and compute the DFT value of $x(n) = 0.5^n$; $0 \leq n \leq 3$

Solution:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j(\frac{2\pi}{N})kn} = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^{N-1} (a^n W_N^{kn}) = \sum_{n=0}^{N-1} [aW_N^k]^n = \frac{1 - a^N W_N^{kN}}{1 - aW_N^k} = \frac{1 - a^N}{1 - aW_N^k}$$

$x(n) = 0.5^n$ Here $a = 0.5$, $N = 4$

$$X(k) = \frac{1 - a^N}{1 - aW_N^k} = \frac{1 - 0.5^4}{1 - (0.5)W_4^k} = \frac{0.9375}{1 - (0.5)e^{-j\left(\frac{2\pi k}{4}\right)}}$$

$$(iv) x(n) = e^{j\frac{2\pi k_0 n}{N}}$$

$$\text{Solution: } X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi}{N}\right)kn} = \sum_{n=0}^{N-1} e^{j\frac{2\pi k_0 n}{N}} e^{-j\left(\frac{2\pi}{N}\right)kn} = \sum_{n=0}^{N-1} e^{-j\frac{2\pi(k-k_0)n}{N}} = \frac{1 - e^{-j2\pi(k-k_0)}}{1 - e^{-j\frac{2\pi}{N}(k-k_0)}}$$

$$X(k) = \frac{1 - e^{-j2\pi(k-k_0)}}{1 - e^{-j\frac{2\pi}{N}(k-k_0)}} = 0 \text{ for } k \neq k_0 \quad \text{and for } k = k_0; X(k) = \sum_{n=0}^{N-1} 1 = N\delta(k - k_0)$$

$$(v) x(n) = \begin{cases} 1; & 0 \leq n \leq \left(\frac{N}{2} - 1\right) \\ 0; & \frac{N}{2} \leq n \leq (N - 1) \end{cases} \quad (N \text{ even})$$

Solution:

$$X(k) = \sum_{n=0}^{N/2-1} x(n) e^{-j\left(\frac{2\pi}{N}\right)kn} = \sum_{n=0}^{N/2-1} 1 e^{-j\left(\frac{2\pi}{N}\right)kn} = \frac{1 - e^{-j\left(\frac{2\pi k}{N}\right)\frac{N}{2}}}{1 - e^{-j\left(\frac{2\pi k}{N}\right)}} = \frac{1 - e^{-j\pi k}}{1 - e^{-j\left(\frac{2\pi k}{N}\right)}} = \frac{1 - (-1)^k}{1 - e^{-j\left(\frac{2\pi k}{N}\right)}}$$

$$(vi) x(n) = \cos\left(\frac{2\pi k_0 n}{N}\right)$$

$$\text{Solution: } x(n) = \cos\left(\frac{2\pi k_0 n}{N}\right) = \frac{e^{j\frac{2\pi k_0 n}{N}} + e^{-j\frac{2\pi k_0 n}{N}}}{2} = \frac{W_N^{-k_0 n} + W_N^{k_0 n}}{2} = \frac{W_N^{k_0 n} + W_N^{-k_0 n}}{2}$$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^{N-1} \left(\frac{W_N^{k_0 n} + W_N^{-k_0 n}}{2} \right) W_N^{kn} = \sum_{n=0}^{N-1} \left(\frac{W_N^{(k+k_0)n} + W_N^{(k-k_0)n}}{2} \right)$$

$$X(k) = \left[\frac{N}{2} [\delta((k - k_0)) + \delta((k + k_0))] \right]$$

$$(vii) x(n) = \sin\left(\frac{2\pi k_0 n}{N}\right)$$

$$\text{Solution: } x(n) = \sin\left(\frac{2\pi k_0 n}{N}\right) = \frac{e^{j\frac{2\pi k_0 n}{N}} - e^{-j\frac{2\pi k_0 n}{N}}}{2j} = \frac{W_N^{-k_0 n} - W_N^{k_0 n}}{2j} = \frac{W_N^{k_0 n} - W_N^{-k_0 n}}{2j}$$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^{N-1} \left(\frac{W_N^{k_0 n} - W_N^{-k_0 n}}{2j} \right) W_N^{kn} = \sum_{n=0}^{N-1} \left(\frac{W_N^{(k+k_0)n} - W_N^{(k-k_0)n}}{2j} \right)$$

$$X(k) = X(k) = \left[\frac{N}{2} [\delta((k - k_0)) - \delta((k + k_0))] \right]$$

3. Compute the N point DFT of $x(n) = \begin{cases} \frac{1}{3}; & 0 \leq n \leq 2 \\ 0; & \text{otherwise} \end{cases}$

Solution: Here $N = 3$ and $x(n) = \{0.33, 0.33, 0.33\}$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi nk}{N}} = \sum_{n=0}^2 x(n) e^{-j\frac{2\pi nk}{3}} = 0.33 + 0.33e^{-j\frac{2\pi k}{3}} + 0.33e^{-j\frac{2\pi 2k}{3}}$$

$$X(0) = 0.33 + 0.33 + 0.33 = 1$$

$$X(1) = 0.33 + 0.33e^{-j\frac{2\pi}{3}} + 0.33e^{-j\frac{4\pi}{3}} = 0.33 - 0.165 - j0.286 - 0.165 + j0.286 = 0$$

$$X(2) = 0.33 + 0.33e^{-j\frac{4\pi}{3}} + 0.33e^{-j\frac{8\pi}{3}} = 0.33 - 0.165 + j0.286 - 0.165 - j0.286 = 0$$

4. Find the DFT of the sequence $x(n) = \{0, 1, 2, 3\}$. Plot the magnitude and phase spectrum

Solution: $N = 4$

$$X(K) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi nk}{N}} = \sum_{n=0}^3 x(n) e^{-j\frac{2\pi nk}{4}} = \sum_{n=0}^3 x(n) W_4^{kn}$$

$$X(K) = x(0)e^0 + x(1)e^{-j\frac{2\pi k}{4}} + x(2)e^{-j\frac{4\pi k}{4}} + x(3)e^{-j\frac{6\pi k}{4}}$$

$$X(K) = 0 + 1e^{-j\frac{\pi k}{2}} + 2e^{-j\pi k} + 3e^{-j\frac{6\pi k}{4}} = 1e^{-j\frac{\pi k}{2}} + 2e^{-j\pi k} + 3e^{-j\frac{6\pi k}{4}}$$

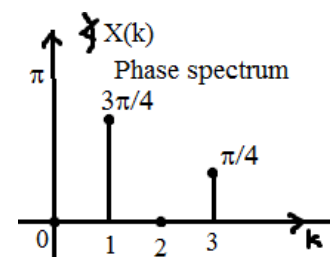
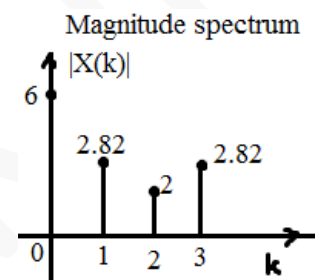
$$X(0) = (1)e^0 + (2)e^0 + (3)e^0 = 6$$

$$X(1) = (1)e^{-j\frac{\pi}{2}} + (2)e^{-j\pi} + (3)e^{-j\frac{6\pi}{4}} = -j - 2 + 3j = -2 + 2j$$

$$X(2) = (1)e^{-j\pi} + (2)e^{-j2\pi} + (3)e^{-j3\pi} = -1 + 2 - 3 = -2$$

$$X(3) = (1)e^{-j\frac{3\pi}{2}} + (2)e^{-j3\pi} + (3)e^{-j\frac{18\pi}{4}} = j - 2 - 3j = -2 - 2j$$

$$X(k) = \{6, (-2 + 2j), -2, (-2 - 2j)\}$$



5. Find the DFT of the sequence $x(n) = \begin{pmatrix} 1; & 0 \leq n \leq 2 \\ 0; & \text{otherwise} \end{pmatrix}$ for (i) $N = 4$ (ii) $N = 8$. Plot magnitude spectrum of $X(K)$ and comment on the result obtained.

Solution: (i) Method 1: Here $N = 4$ and $x(n) = \{1, 1, 1\}$

Modify the sequence $x(n)$ for a length of 4 by padding a zero $x(n) = \{1, 1, 1, 0\}$

$$X(K) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi nk}{N}} = \sum_{n=0}^3 x(n) e^{-j\frac{2\pi nk}{4}} = \sum_{n=0}^3 x(n) W_4^{kn}$$

$$X(K) = x(0)e^0 + x(1)e^{-j\frac{2\pi k}{4}} + x(2)e^{-j\frac{4\pi k}{4}} + x(3)e^{-j\frac{6\pi k}{4}} = 1(1) + (1)e^{-j\frac{\pi k}{2}} + (1)e^{-j\pi k} + (0)e^{-j\frac{6\pi k}{4}}$$

$$X(0) = 1(1) + (1)e^0 + (1)e^0 + (0)e^0 = 3$$

$$X(1) = 1 + (1)e^{-j\frac{\pi}{2}} + (1)e^{-j\pi} + (0)e^{-j\frac{6\pi}{4}} = 1 - j - 1 = 0 = -j$$

$$X(2) = 1 + (1)e^{-j\pi} + (1)e^{-j2\pi} + (0)e^{-j3\pi} = 1 - 1 + 1 + 0 = 1$$

$$X(3) = 1 + (1)e^{-j\frac{\pi 3}{2}} + (1)e^{-j\pi 3} + (0)e^{-j\frac{18\pi}{4}} = 1 + j - 1 + 0 = j$$

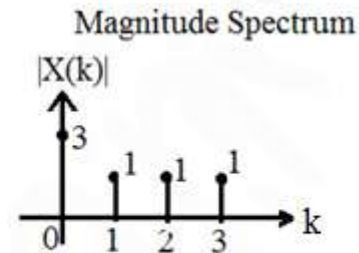
$$X(k) = \{1, -j, 1, j\}$$

Method 2: Here $N = 4$ and $x(n) = \{1, 1, 1\}$

Modify the sequence $x(n)$ for a length of 4 by padding a zero $x(n) = \{1, 1, 1, 0\}$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi nk}{N}} = \sum_{n=0}^3 x(n) e^{-j\frac{2\pi nk}{4}} = \sum_{n=0}^3 x(n) W_4^{kn}$$

$$X(K) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ -j \\ 1 \\ j \end{bmatrix}$$



Procedure for calculating N= 4 point DFT using calculator (complex mode and degrees)

enter the values as shown

$$(1|0A) + (1|-90A) + (1|-180A) + (0|-270A)$$

Press Calculate 0 = to get $X(0)$; Press Calculate 1 = to get $X(1)$; Press Calculate 2 = to get $X(2)$;

Press Calculate 3 = to get $X(3)$;

(ii) 8 point DFT $N = 8$ and $x(n) = \{1, 1, 1\}$

Modify the sequence $x(n)$ for a length of 8 by padding a zeros $x(n) = \{1, 1, 1, 0, 0, 0, 0, 0\}$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi nk}{N}} = \sum_{n=0}^7 x(n) e^{-j\frac{2\pi nk}{8}} = \sum_{n=0}^7 x(n) W_8^{kn}$$

$$k = 0; \quad X(0) = \sum_{n=0}^7 x(n) W_8^{0n} = x(0) + x(1) + x(2) + x(3) + x(4) + x(5) + x(6) + x(7)$$

$$X(0) = 1 + 1 + 1 + 0 + 0 + 0 + 0 + 0 = 3$$

$$k = 1; \quad X(1) = \sum_{n=0}^7 x(n) W_8^{1n} = x(0) + x(1)W_8^1 + x(2)W_8^2 = 1 + 1(0.707 - j0.707) + 1(-j)$$

$$X(1) = 1.707 - j1.707 \quad (\text{only first 3 samples are considered as the other sample values are zero})$$

$$k = 2; \quad X(2) = \sum_{n=0}^7 x(n) W_8^{2n} = x(0) + x(1)W_8^2 + x(2)W_8^4 = 1 + 1(-j) + 1(-1) = -j$$

$$k = 3; \quad X(3) = \sum_{n=0}^7 x(n) W_8^{3n} = x(0) + x(1)W_8^3 + x(2)W_8^6$$

$$X(3) = 1 + 1(-0.707 - j0.707) + 1(j) = 0.293 + j0.293$$

$$X(k) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -1 & -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & j & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & j & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -j & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & j & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & j & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -1 & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -j & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 8 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

7. Find the IDFT of X(k) given

$$X(k) = \{255, 48.63 + j166.05, -51 + j102, -78.63 + j46.05, -85 - 78.63 - j46.05, -51 - j102, 48.63 - j166.05\}$$

$$\text{Solution: } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} = \frac{1}{8} \sum_{k=0}^7 X(k) W_8^{-kn}$$

$$x(n) = \frac{1}{8} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & j & -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -1 & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -j & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -j & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & j & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & j & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -1 & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & \frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -j & -\frac{1}{\sqrt{2}} - \frac{j}{\sqrt{2}} & -1 & -\frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} & j & \frac{1}{\sqrt{2}} + \frac{j}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 255 \\ 48.63 + j166.05 \\ -51 + j102 \\ -78.63 + j46.05 \\ -85 \\ -78.63 - j46.05 \\ -51 - j102 \\ 48.63 - j166.05 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{bmatrix}$$

Procedure for calculating IDFT for N=8 using calculator (complex mode and degrees)

enter the values as shown. Angles should be positive

$$(1/8) ((255|0A) + ((48.63+j166.05)*1|45A) + ((-51+j102)*1|90A) + ((-78+j46.05)*1|135A) + ((-85)*1|180A) + ((-78.63-j46.05)*1|225A) + ((-51-j102)*1|270A) + ((48.63-j166.05)*1|315A))$$

Press Calculate 0 = to get x(0); Press Calculate 1 = to get x(1); Press Calculate 2 = to get x(2);

Proceed till Press Calculate 7 = to get x(7);

8. Find the IDFT of X(k) = { 4, -2j, 0, 2j }

$$\text{Solution: } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W_N^{-kn} = \frac{1}{4} \sum_{k=0}^3 X(k) W_4^{-kn}$$

$$x(n) = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 4 \\ -2j \\ 0 \\ 2j \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

$$x(n) = \{1, 2, 1, 0\}$$

Procedure for calculating IDFT for N=4 using calculator (complex mode and degrees)

enter the values as shown. Angles should be positive

$$(1/4) ((4|0A) + ((-2j)*1|90A) + (0|180A) + (2j)*1|270A)$$

Press Calculate 0 = to get x(0); Press Calculate 1 = to get x(1); Press Calculate 2 = to get x(2);

Press Calculate 3 = to get x(3);

9. Compute the DFT of the sequence $x(n) = \{0, 1, 2, 3\}$ plot the magnitude and phase spectrum.

Solution: Here $N = 4$ and $x(n) = \{0, 1, 2, 3\}$

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi nk}{N}} = \sum_{n=0}^3 x(n) e^{-j\frac{2\pi nk}{4}} = x(0) + x(1)e^{-j\frac{2\pi k}{4}} + x(2)e^{-j\frac{4\pi k}{4}} + x(3)e^{-j\frac{6\pi k}{4}}$$

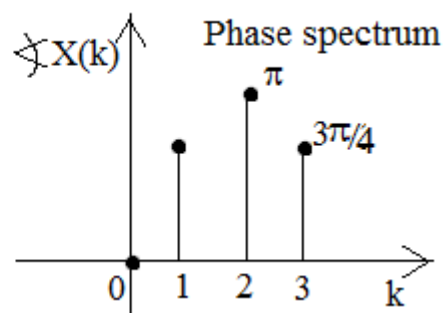
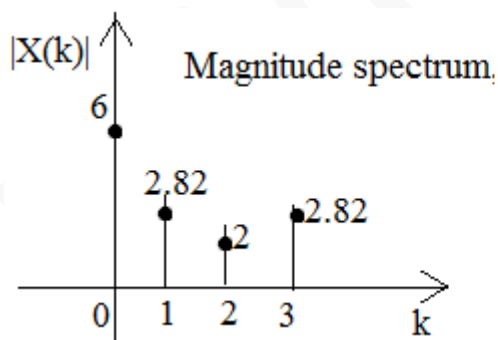
$$X(k) = 0 + (1)e^{-j\frac{\pi k}{2}} + (2)e^{-j\pi k} + x(3)e^{-j\frac{3\pi k}{2}}$$

$$k = 0; X(0) = 0 + (1)e^{-j0} + (2)e^{-j0} + (3)e^{-j0} = 1 + 2 + 3 = 6$$

$$k = 1; X(1) = 0 + (1)e^{-j\frac{\pi}{2}} + (2)e^{-j\pi} + (3)e^{-j\frac{3\pi}{2}} = -j + (2)(-1) + 3(j) = -2 + 2j$$

$$k = 2; X(2) = 0 + (1)e^{-j\pi} + (2)e^{-j2\pi} + x(3)e^{-j3\pi} = -1 + 2 - 3 = -2$$

$$k = 3; X(3) = 0 + (1)e^{-j\frac{3\pi}{2}} + (2)e^{-j3\pi} + x(3)e^{-j\frac{9\pi}{2}} = j - 2 - 3j = -2 - 2j$$



10. Find the 8 point DFT of $x(n) = \delta(n) + \delta(n-1) + \delta(n-2)$. Plot the magnitude and phase spectrum.

Solution: $x(n) = \{1, 1, 1\}$ since $N=8$, $x(n) = \{1, 1, 1, 0, 0, 0, 0, 0\}$

Refer problem 5 (ii)

12. Given $X(k) = \{2, 1+j, 0, 1-j\}$ find $x(n)$

Solution: Method 1: Here $N = 4$ and $X(k) = \{ 2, 1 + j, 0, 1 - j \}$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi nk}{N}} = \frac{1}{4} \sum_{k=0}^3 X(k) e^{j \frac{2\pi nk}{4}} = \frac{1}{4} \left(X(0) + X(1) e^{j \frac{2\pi n}{4}} + X(2) e^{j \frac{4\pi n}{4}} + X(3) e^{j \frac{6\pi n}{4}} \right)$$

$$x(n) = \frac{1}{4} \left(2 + (1 + j) e^{j \frac{2\pi n}{4}} + (0) e^{j \frac{4\pi n}{4}} + (1 - j) e^{j \frac{6\pi n}{4}} \right) = \frac{1}{4} \left(2 + (1 + j) e^{j \frac{\pi n}{2}} + (1 - j) e^{j \frac{3\pi n}{2}} \right)$$

$$n = 0; \quad x(0) = \frac{1}{4} (2 + (1 + j) e^0 + (1 - j) e^0) = \frac{1}{4} (2 + (1 + j) + (1 - j)) = 1$$

$$n = 1; \quad x(1) = \frac{1}{4} \left(2 + (1 + j) e^{j \frac{\pi}{2}} + (1 - j) e^{j \frac{3\pi}{2}} \right) = \frac{1}{4} (2 + (1 + j)(j) + (1 - j)(-j)) = 0$$

$$n = 2; \quad x(2) = \frac{1}{4} \left(2 + (1 + j) e^{j \frac{\pi 2}{2}} + (1 - j) e^{j \frac{3\pi 2}{2}} \right) = \frac{1}{4} (2 - 1 - j - 1 + j) = 0$$

$$n = 3; \quad x(3) = \frac{1}{4} \left(2 + (1 + j) e^{j \frac{\pi 3}{2}} + (1 - j) e^{j \frac{9\pi}{2}} \right) = \frac{1}{4} (2 + (1 + j)(-j) + (1 - j)(j)) = 1$$

$$x(n) = \{ 1, 0, 0, 1 \}$$

Method 2: Here $N = 4$ and $X(k) = \{ 2, 1 + j, 0, 1 - j \}$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j \frac{2\pi nk}{N}} = \frac{1}{4} \sum_{k=0}^3 X(k) e^{j \frac{2\pi nk}{4}} = \frac{1}{4} \sum_{k=0}^3 X(k) W_4^{-kn}$$

$$x(n) = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 2 \\ 1+j \\ 0 \\ 1-j \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

13. Find the 16 point IDFT of $X(k) = \cos\left(\frac{6\pi k}{16}\right) + j3 \sin\left(\frac{10\pi k}{16}\right)$

$$\text{Solution: } X(k) = \cos\left(\frac{6\pi k}{16}\right) + j3 \sin\left(\frac{10\pi k}{16}\right) = \cos\left(\frac{2\pi(3k)}{16}\right) + j3 \sin\left(\frac{2\pi(5k)}{16}\right)$$

$$X(k) = \frac{e^{j\left(\frac{2\pi}{16}\right)3k} + e^{-j\left(\frac{2\pi}{16}\right)3k}}{2} + 3 \left(\frac{e^{j\left(\frac{2\pi}{16}\right)5k} - e^{-j\left(\frac{2\pi}{16}\right)5k}}{2j} \right) = \frac{W_{16}^{-3k} + W_{16}^{3k}}{2} + 3j \left(\frac{W_{16}^{-5k} - W_{16}^{5k}}{2j} \right)$$

$$x(n) = \frac{1}{16} \sum_{k=0}^{15} x(n) W_{16}^{-kn} = \frac{1}{16} \left[16 \left[\frac{\delta(n+3) + \delta(n-3)}{2} \right] \right] + \frac{3}{16} \left[16 \left[\frac{\delta(n+5) - \delta(n-5)}{2} \right] \right]$$

For a casual sequence; $\delta(n+3) = \delta(n+5) = 0$

$$\text{Hence } x(n) = \begin{cases} \left(\frac{1}{2}\right) & n = 3 \\ \left(\frac{3}{2}\right) & n = 5 \\ 0 & \text{else where} \end{cases}$$

14. Find the 10 point IDFT of $X(k) = \begin{cases} 3; & k = 0 \\ 1; & 1 \leq k \leq 9 \end{cases}$

Solution: $X(k) = 1 + 2\delta(k) = 1 + \left(\frac{1}{5}\right)10 \delta(k)$

Taking IDFT on both sides

$$x(n) = \delta(n) + \left(\frac{1}{5}\right) = \left\{ \left(\frac{6}{5}\right), \left(\frac{1}{5}\right), \left(\frac{1}{5}\right), \left(\frac{1}{5}\right), \left(\frac{1}{5}\right), \left(\frac{1}{5}\right), \left(\frac{1}{5}\right), \left(\frac{1}{5}\right), \left(\frac{1}{5}\right), \left(\frac{1}{5}\right) \right\}$$

Note

$$\begin{aligned} N\delta(k) &\leftarrow \text{IDFT}_N \rightarrow 1 \\ 1 &\leftarrow \text{IDFT}_N \rightarrow \delta(n) \end{aligned}$$

15. Find the 5 point of the discrete time signal $x(n) = \{ 1, 2, 3, 1 \}$

Solution: $N = 5$; hence $x(n) = \{ 1, 2, 3, 1, 0 \}$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^4 x(n) W_5^{kn} = x(1)W_5^0 + x(1)W_5^k + x(2)W_5^{2k} + x(3)W_5^{3k} + x(4)W_5^{4k}$$

$$X(k) = 1 + 2W_5^k + 3W_5^{2k} + W_5^{3k}$$

$$X(0) = 1 + 2W_5^0 + 3W_5^0 + W_5^0 = 7$$

$$X(1) = 1 + 2W_5^1 + 3W_5^2 + W_5^3 = -1.618 - j3.076$$

$$X(2) = 1 + 2W_5^2 + 3W_5^4 + W_5^6 = 1 + 2W_5^2 + 3W_5^4 + W_5^1 = 0.618 + j0.712$$

$$X(3) = 1 + 2W_5^3 + 3W_5^6 + W_5^9 = 1 + 2W_5^3 + 3W_5^1 + W_5^4 = 0.618 - j0.712$$

$$X(4) = 1 + 2W_5^4 + 3W_5^8 + W_5^{12} = 1 + 2W_5^4 + 3W_5^3 + W_5^2 = -1.618 + j3.076$$

$$\begin{aligned} W_5^0 &= 1 \\ W_5^1 &= e^{-j2\pi/5} = 0.309 - j0.951 \\ W_5^2 &= e^{-j4\pi/5} = -0.809 - j0.587 \\ W_5^3 &= e^{-j6\pi/5} = -0.809 + j0.587 \\ W_5^4 &= e^{-j8\pi/5} = 0.309 + j0.951 \end{aligned}$$

16. Find the DFT of $x(n) = u(n) - u(n-5)$

Solution: $x(n) = \{ 1, 1, 1, 1, 1 \}$ $N = 5$

Refer previous problem to find out DFT $X(k) = \{ 5, 0, 0, 0, 0 \}$

17. Find the 8 point DFT of $x(n) = \{ 2, 2, 2, 2, 2, 2, 2, 2 \}$

Solution: $x(n) = 2[1, 1, 1, 1, 1, 1, 1, 1] = 2(1)^n$ $0 \leq n \leq 7$ hence $N = 8$

Taking DFT on both sides; $X(k) = 2 N \delta(k) = 16 \delta(k)$

18. Find the DFT of $x(n) = (-1)^n$; $0 \leq n \leq 3$

Solution: $x(n) = (-1)^n = \{1, -1, 1, -1\}$ Use 4 point Twiddle matrix and determine DFT

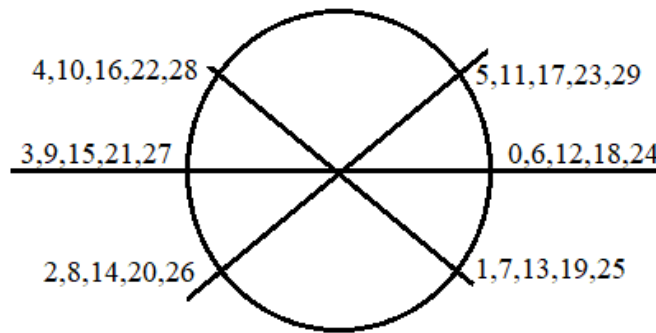
$$X(k) = \{0, 0, 4, 0\}$$

19. Find the DFT of $x(n) = (-1)^{n+1} : 0 \leq n \leq 5$

Solution: $x(n) = (-1)^{n+1} = \{-1, 1, -1, 1, -1, 1\}$ $N = 6$

$$X(k) = \sum_{n=0}^5 x(n) W_6^{kn} = x(0)W_6^0 + x(1)W_6^k + x(2)W_6^{2k} + x(3)W_6^{3k} + x(4)W_6^{4k} + x(5)W_6^{5k}$$

$$\begin{aligned} W_6^0 &= 1 \\ W_6^1 &= 0.5 - j1.732 \\ W_6^2 &= -0.5 + j1.732 \\ W_6^3 &= -1 \\ W_6^4 &= -0.5 - j1.732 \\ W_6^5 &= 0.5 + j1.732 \end{aligned}$$



$$X(0) = -1 + 1 - 1 + 1 - 1 + 1 = 0$$

$$X(1) = x(0)W_6^0 + x(1)W_6^1 + x(2)W_6^2 + x(3)W_6^3 + x(4)W_6^4 + x(5)W_6^5$$

$$X(1) = -1 + 1(0.5 - j1.732) - 1(-0.5 + j1.732) + 1(-1) + (-1)(-0.5 - j1.732) + 1(0.5 + j1.732) = 0$$

$$X(2) = x(0)W_6^0 + x(1)W_6^2 + x(2)W_6^4 + x(3)W_6^6 + x(4)W_6^8 + x(5)W_6^{10}$$

$$X(2) = -1 + 1(-0.5 + j1.732) - 1(-0.5 - j1.732) + 1(1) - 1(0.5 - j1.732) + 1(-0.5 + j1.732) = 0$$

$$X(3) = x(0)W_6^0 + x(1)W_6^3 + x(2)W_6^6 + x(3)W_6^9 + x(4)W_6^{12} + x(5)W_6^{15}$$

$$X(3) = -1 + 1(-1) - 1(1) + 1(-1) - 1(1) + 1(-1) = -6$$

$$X(4) = x(0)W_6^0 + x(1)W_6^4 + x(2)W_6^8 + x(3)W_6^{12} + x(4)W_6^{16} + x(5)W_6^{20}$$

$$X(4) = x(0)W_6^0 + x(1)W_6^4 + x(2)W_6^2 + x(3)W_6^0 + x(4)W_6^4 + x(5)W_6^2$$

$$X(4) = -1 + 1(-0.5 + j1.732) - 1(-0.5 - j1.732) + 1(1) - 1(-0.5 + j1.732) + 1(-0.5 - j1.732) = 0$$

$$X(5) = x(0)W_6^0 + x(1)W_6^5 + x(2)W_6^{10} + x(3)W_6^{15} + x(4)W_6^{20} + x(5)W_6^{25}$$

$$X(5) = x(0)W_6^0 + x(1)W_6^5 + x(2)W_6^4 + x(3)W_6^3 + x(4)W_6^2 + x(5)W_6^1$$

$$X(5) = -1 + 1(0.5 + j1.732) - 1(-0.5 + j1.732) + 1(-1) - 1(-0.5 - j1.732) + 1(0.5 - j1.732) = 0$$

$$X(k) = \{0, 0, 0, -6, 0, 0\}$$

20. Find the DFT of $x(n) = (-1)^{n+1} : 0 \leq n \leq 5$

$$\text{Solution: } X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{n=0}^{N-1} (-1)^n W_N^{kn} = \sum_{n=0}^{N-1} (-1 W_N^k)^n = \frac{1 - (-1)^N W_N^{kN}}{1 - (-1) W_N^k} = \frac{1 - (-1)^N}{1 + W_N^k}$$

$$\text{when } N \text{ is odd ; } X(k) = \frac{1 - (-1)^N}{1 + W_N^k} = \frac{2}{1 + W_N^k}$$

$$\text{when } N \text{ is even; } k = \frac{N}{2} \text{ hence } W_N^k = W_N^{\frac{N}{2}} = e^{-j\frac{2\pi}{N}(\frac{N}{2})} = e^{-j\pi} = -1$$

$$X(k) = \sum_{n=0}^{N-1} (-1)^n (-1)^n \sum_{n=0}^{N-1} (1)^n = [N]_{k=\frac{N}{2}} = N\delta\left(k - \frac{N}{2}\right)$$

Ex: $x(n) = \{1, -1, 1, -1\}$ for $N = 4$ and its DFT is $X(k) = \{0, 0, 4, 0\}$

Properties of DFT

The signal $x(n)$ and its DFT $X(k)$ is represented as a pair $x(n) \leftarrow \text{DFT} \rightarrow X(k)$

Periodicity: If $x(n)$ and $X(k)$ are DFT pair,

If $x(n + N) = x(n)$ for all n

then $X(k + N) = X(k)$ for all k

Proof: By the definition of DFT

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)}$$

put $k = k + N$, then

$$X(k + N) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi n(k+N)}{N}\right)} = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi nk}{N}\right)} \cdot e^{-j\left(\frac{2\pi nN}{N}\right)} = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi nk}{N}\right)} (1) = X(k)$$

22. Consider two periodic sequences $x(n)$ and $y(n)$ with period N and M respectively. The sequence $w(n)$ is defined as $w(n) = x(n) + y(n)$. Show that $w(n)$ is periodic with period MN (ii) Also show that $W(k)$ represents MN point DFT of an MN length sequence $w(n)$

Solution: (i) The sequence $w(n)$ is periodic with period MN , hence

$$w(n + MN) = x(n + MN) + y(n + MN)$$

$$x(n) = x(n+N) = x(n+2N) = x(n+3N) = \dots = x(n + MN) \quad \text{for all integer multiples of } N$$

$$\text{Similarly, } y(n) = y(n+M) = y(n+2M) = y(n+3M) = \dots = y(n + MN) \quad \text{for all integer multiples of } M$$

$$w(n + MN) = x(n) + y(n) = w(n). \text{ Hence } w(n) \text{ is periodic with period } MN$$

(ii) $w(n)$ is a sequence of length MN and its MN point DFT is given as

$$W(k) = \sum_{n=0}^{MN-1} w(n) e^{-j\left(\frac{2\pi kn}{MN}\right)} \quad k = 0, 1, 2, \dots, (MN - 1)$$

Linearity:

If $x_1(n) \leftarrow \text{DFT} \rightarrow X_1(k)$ and $x_2(n) \leftarrow \text{DFT} \rightarrow X_2(k)$, then

$a x_1(n) + b x_2(n) \leftarrow \text{DFT} \rightarrow a X_1(k) + b X_2(k)$ where a and b are constants.

Proof: By the definition of DFT

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)} \quad \text{if } x(n) = a x_1(n) + b x_2(n), \text{ then}$$

$$X(k) = \sum_{n=0}^{N-1} [a x_1(n) + b x_2(n)] e^{-j\left(\frac{2\pi kn}{N}\right)} = \sum_{n=0}^{N-1} a x_1(n) e^{-j\left(\frac{2\pi kn}{N}\right)} + \sum_{n=0}^{N-1} b x_2(n) e^{-j\left(\frac{2\pi kn}{N}\right)} = a X_1(k) + b X_2(k)$$

23. Determine the 4 point DFT of the sequence $x(n) = \cos\left(\frac{\pi n}{4}\right) + \sin\left(\frac{\pi n}{4}\right)$ using linearity property

Solution: Given $N = 4$, Let $x_1(n) = \cos\left(\frac{\pi n}{4}\right) = \left\{1, \frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right\}$ and

$$x_2(n) = \sin\left(\frac{\pi n}{4}\right) = \left\{0, \frac{1}{\sqrt{2}}, 1, \frac{1}{\sqrt{2}}\right\}$$

$$X_1(K) = \sum_{n=0}^{N-1} x_1(n) W_4^{kn} = \sum_{n=0}^3 x_1(n) W_4^{kn}$$

$$X_2(K) = \sum_{n=0}^{N-1} x_2(n) W_4^{kn} = \sum_{n=0}^3 x_2(n) W_4^{kn}$$

$$X_1(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 0.707 \\ 1 \\ -0.707 \end{bmatrix} = \begin{bmatrix} 1 \\ 1-j1.414 \\ 1 \\ 1+j1.414 \end{bmatrix}$$

$$X_2(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 0.707 \\ 1 \\ 0.707 \end{bmatrix} = \begin{bmatrix} 2.414 \\ -1 \\ -0.414 \\ -1 \end{bmatrix}$$

$$X(k) = X_1(k) + X_2(k) = \{X_1(0) + X_2(0), X_1(1) + X_2(1), X_1(2) + X_2(2), X_1(3) + X_2(3)\}$$

$$X(k) = \{3.414, -j1.414, 0.586, j1.414\}$$

1.5(c) - Circular Symmetries of a Sequence

If $X(k)$ represents the N – point DFT of sequence $x(n)$, then the IDFT $X(k)$ yield a periodic sequence $x_p(n)$ which is related to original sequence $x(n)$ as

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n - lN) \quad \text{where } l \text{ is the index}$$

$x_p(n)$ is the periodic repetition of $x(n)$ and hence it can be concluded that

$$x(n) \leftarrow \text{DFT} \rightarrow X(k) \quad x_p(n) \leftarrow \text{DFT} \rightarrow X(k) \quad \text{and} \quad x(n) = \begin{cases} x_p(n) & \text{for } 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

If $x_p(n)$ is shifted by k units to right and the new sequence $x_p^{\#}(n)$ can be expressed as

$$x_p^{\#}(n) = x_p(n - k) = \sum_{l=-\infty}^{\infty} x(n - k - lN)$$

$$\text{The corresponding finite duration sequence } x^{\#}(n) = \begin{cases} x_p^{\#}(n) & \text{for } 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

The sequence $x^{\#}(n)$ related to $x(n)$ by circular shift and represented as

$$x^{\#}(n) = \sum_{l=0}^{N-1} x(n - k - lN) = x(n - k, \text{Modulo } N) = x((n - k))_N$$

Illustration: Let the shift $k = 2$ and periodicity $N = 4$

$$x^{\#}(n) = x((n - k))_N = x((n - 2))_4$$

$x^{\#}(0) = x((-2))_4 = x(2)$; Shift $x(n)$ circularly two samples in anticlockwise direction

$$x^{\#}(1) = x((-1))_4 = x(3)$$
; Shift $x(n)$ circularly one sample in anticlockwise direction

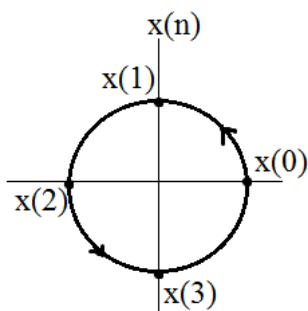
Note: If the argument n is in between 0 and $N - 1$, then leave it as it is; otherwise add/subtract multiples of N from n until the argument lies in between 0 and $N - 1$.

$$x^{\#}(2) = x((0))_4 = x(0);$$

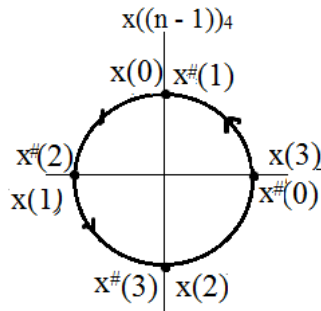
No shift or shift $x(n)$ by 4 samples in ACW direction

$$x^{\#}(3) = x((1))_4 = x(1);$$

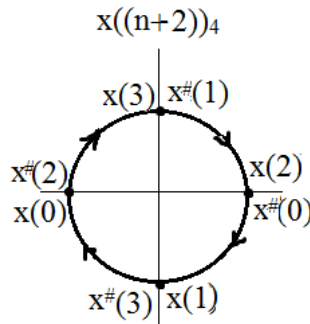
Shift $x(n)$ circularly one sample in clockwise direction



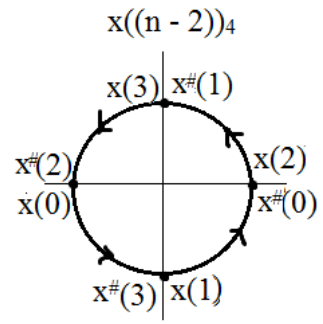
Plot $x(n)$ in ACW direction



Shift $x(n)$ in ACW direction one sample



Shift $x(n)$ in CW direction two samples



Shift $x(n)$ in ACW direction two samples

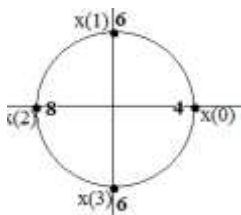
Circularly Even Sequence:

A sequence is said to be circularly even if it is symmetric about the point zero on the circle. $x(N - n) = x_1(n)$

Illustration:

consider $x(n) = \{ 4, 6, 8, 6 \}$; $N = 4$ $x_1(n) = x(4 - n)$
 $x_1(0) = x(4 - 0) = x(0)$; $x_1(1) = x(4 - 1) = x(3)$;
 $x_1(2) = x(4 - 2) = x(2)$; $x_1(3) = x(4 - 3) = x(1)$

$$x_1(n) = \{ 4, 6, 8, 6 \}$$



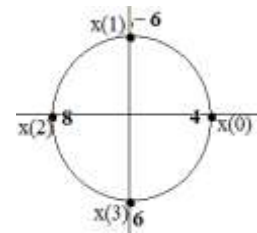
Circularly Odd Sequence:

A sequence is said to be circularly odd if it is anti-symmetric about the point zero on the circle. $x(N - n) = -x_1(n)$

Illustration:

consider $x(n) = \{ 4, -6, 8, 6 \}$; $N = 4$ $x_1(n) = x(4 - n)$
 $x_1(0) = x(4 - 0) = x(0)$; $x_1(1) = x(4 - 1) = x(3)$;
 $x_1(2) = x(4 - 2) = x(2)$; $x_1(3) = x(4 - 3) = x(1)$

$$x_1(n) = \{ 4, 6, 8, -6 \}$$



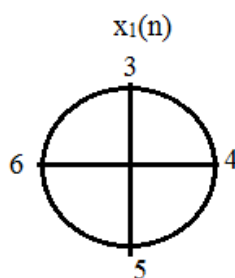
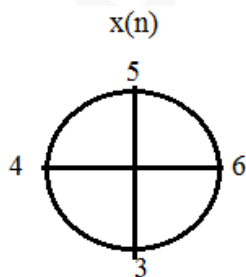
Circularly Folded Sequence : A circularly folded sequence is represented by $x((-n))_N$ and is obtained by plotting $x(n)$ clockwise direction along the circle. It is represented as $x((-n))_N = x(N - n)$

24. If $x(n)$ is a finite duration sequence of length N , show that $x((-n))_N = X((N - n))_N$

Solution: If N is the period and if $x(-n)$ is periodic with N , then $x((-n))_N = x(-n + qN)$ $-\infty \leq q \leq \infty$

Let $q = 1$, then $x((-n))_N = x(-n + N) = x((N - n))_N$

25. If $x(n) = \{ 6, 5, 4, 3 \}$ Sketch $x_1(n) = x((n - 2))_4$ $0 \leq n \leq 3$



shift $x(n)$ 2 samples in ACW direction
 $x_1(n) = \{ 4, 3, 6, 5 \}$

26. If $x(n) = \{ 1, 2, 1, -1, 2, 3 \}$ Determine (i) $x((n - 3))_6$ (ii) $x((n + 2))_6$ and (iii) $x((n - 3))_6$

Solution: (i) Let $x_1(n) = x((n - 3))_6$

$$x_1(0) = x((-3))_6 = x(6-3) = x(3) = -1 ;$$

$$x_1(1) = x((1-3))_6 = x(-2) = x(6-2) = x(4) = 2$$

$$x_1(2) = x((2-3))_6 = x(-1) = x(6-1) = x(5) = 3$$

$$x_1(3) = x((3-3))_6 = x(0) = 1$$

$$x_1(4) = x((4-3))_6 = x(1) = 2$$

$$x_1(5) = x((5-3))_6 = x(2) = 1$$

$$x_1(n) = \{ -1, 2, 3, 1, 2, 1 \} \quad (\text{Shift } x(n) \text{ by three samples in ACW direction})$$

$$(ii) \text{ Let } x_2(n) = x((n+2))_6$$

$$x_2(0) = x((2))_6 = 1 ; \quad x_2(1) = x((1+2))_6 = x(3) = -1 ; \quad x_2(2) = x((2+2))_6 = x(4) = 2$$

$$x_2(3) = x((3+2))_6 = x(5) = 3 ; \quad x_2(4) = x((4+2))_6 = x(6) = x(0) = 1 ; \quad x_2(5) = x((5+2))_6 = x(7) = x(1) = 2$$

$$x_2(n) = \{ 1, -1, 2, 3, 1, 2 \} \quad (\text{Shift } x(n) \text{ by two samples in CW direction})$$

$$(iii) \text{ Let } x_3(n) = x((n+3))_6$$

$$x_3(0) = x((3))_6 = x(3) = -1 ; \quad x_3(1) = x((1+3))_6 = x(4) = 2 ; \quad x_3(2) = x((2+3))_6 = x(5) = 3$$

$$x_3(3) = x((3+3))_6 = x(0) = 1 ; \quad x_3(4) = x((4+3))_6 = x(7) = x(1) = 2 ; \quad x_3(5) = x((5+3))_6 = x(8) = x(2) = 1$$

$$x_3(n) = \{ -1, 2, 3, 1, 2, 1 \} \quad (\text{Shift } x(n) \text{ by three samples in CW direction})$$

Symmetry Property for real valued sequence $x(n)$

If $x(n)$ is real, then $X(N-k) = X^*(k) = X(-k)$

Proof: By the definition of DFT

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi kn}{N}\right)}$$

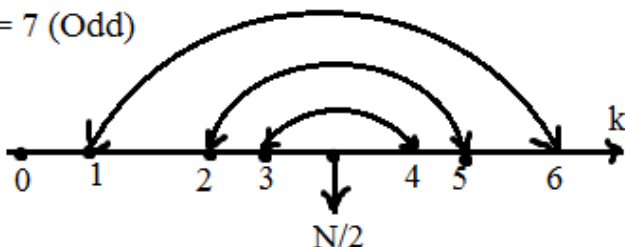
put $k = N-k$

$$X(N-k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi n}{N}(N-k)} = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi nN}{N}} e^{j\frac{2\pi nk}{N}} = X(-k) \quad \text{----- (a)}$$

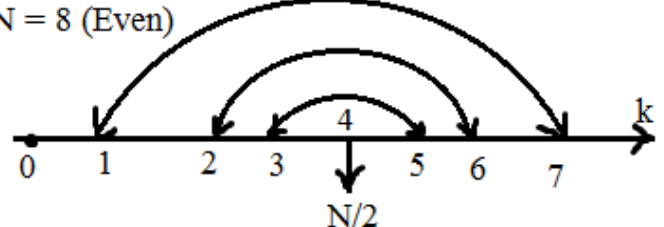
$$X^*(k) = \sum_{n=0}^{N-1} x(n) e^{j\left(\frac{2\pi kn}{N}\right)} \quad \text{----- (b)}$$

From equations (a) and (b) $\Rightarrow X(N-k) = X^*(k) = X(-k)$

$N = 7$ (Odd)



$N = 8$ (Even)



27. The first 5 points of an 8 point DFT of real valued sequence are $\{ 0, 2+j2, -j4, 2-j2, 0 \}$. Find the remaining samples of the DFT.

Solution: Given $N = 8$; The sequence is real and hence symmetry occurs at $k = N/2 = 4$

Using symmetry property of real valued sequence $X(k) = X^*(N-k)$

Put $k = 5$; $X(5) = X^*(N - k) = X^*(8 - 5) = X^*(3) = [2 - j2]^* = 2 + j2$

Put $k = 6$; $X(6) = X^*(N - k) = X^*(8 - 6) = X^*(2) = [-j4]^* = j4$

Put $k = 7$; $X(7) = X^*(N - k) = X^*(8 - 7) = X^*(1) = [2 + j2]^* = 2 - j2$

$X(k) = \{ 0, (2 + j2), -j4, (2 - j2), 0, (2 + j2), j4, (2 - j2) \}$

28. The first 5 values of an 8 point DFT $X(k)$ for a real valued sequence are $\{ 0.25, (0.125 - j0.3018), 0, (0.125 - j0.518), 0 \}$. Find the remaining values. Estimate the value of $x(0)$.

Solution: Given $N = 8$; The sequence is real and hence symmetry occurs at $k = N/2 = 4$

Using symmetry property of real valued sequence $X(k) = X^*(N - k)$

Put $k = 5$; $X(5) = X^*(N - k) = X^*(8 - 5) = X^*(3) = [0.125 - j0.518]^* = 0.125 + j0.518$

Put $k = 6$; $X(6) = X^*(N - k) = X^*(8 - 6) = X^*(2) = [0]^* = 0$

Put $k = 7$; $X(7) = X^*(N - k) = X^*(8 - 7) = X^*(1) = [0.125 - j0.3018]^* = 0.125 + j0.3018$

$X(k) = \{ 0.25, (0.125 - j0.3018), 0, (0.125 - j0.518), 0, (0.125 + j0.518), 0, (0.125 + j0.3018) \}$

Estimation of $x(0)$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j2\pi kn/N} = \frac{1}{8} \sum_{k=0}^7 X(k) e^{j2\pi kn/8}$$

$$\text{put } n = 0; x(0) = \frac{1}{8} [(X(0) + X(1) + X(2) + X(3) + X(4) + X(5) + X(6) + X(7))] = 0.09375$$

29. Compute the 5 point DFT of the sequence $x(n) = \{ 1, 0, 1, 0, 1 \}$ using symmetry property.

Solution: $N = 5$; (odd) symmetry occurs at $k = N/2 = 2.5$.

Hence from symmetry property, $X(4) = X^*(N - k) = X^*(5 - k) = X^*(5 - 4) = X^*(1)$ and $X(3) = X^*(2)$

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{kn} = \sum_{k=0}^5 x(n) W_5^{kn} = x(0) + x(1)W_5^k + x(2)W_5^{2k} + x(3)W_5^{3k} + x(4)W_5^{4k} = 1 + W_5^{2k} + W_5^{4k}$$

Put $k = 0$; $X(0) = 1 + W_5^0 + W_5^0 = 3$

Put $k = 1$; $X(1) = 1 + W_5^2 + W_5^4 = 1 + e^{-j2\pi(2)/5} + e^{-j2\pi(4)/5} = 0.5 + j0.364$

Put $k = 2$; $X(2) = 1 + W_5^{2(2)} + W_5^{4(2)} = 1 + e^{-j2\pi(2)(2)/5} + e^{-j2\pi(4)(2)/5} = 0.5 + j1.538$

From symmetry property, $X(4) = X^*(N - k) = X^*(5 - 4) = X^*(1) = 0.5 - j0.364$

and $X(3) = X^*(N - k) = X^*(5 - 3) = X^*(2) = 0.5 - j1.538$

30 Show that the given sequence $x(n) = 4\delta(n) + 3\delta(n - 1) + 2\delta(n - 2) + \delta(n - 3)$ satisfies symmetry property for $N = 6$

Solution: $x(n) = \{ 4, 3, 2, 1 \}$ modified sequence $x(n) = \{ 4, 3, 2, 1, 0, 0 \}$

$$X(k) = \sum_{n=0}^5 x(n) W_6^{kn} = x(0) + x(1)W_6^k + x(2)W_6^{2k} + x(3)W_6^{3k} + x(4)W_6^{4k} + x(5)W_6^{5k} = 4 + 3W_6^k + 2W_6^{2k} + W_6^{3k}$$

$$X(0) = 4 + 3 + 2 + 1 = 10$$

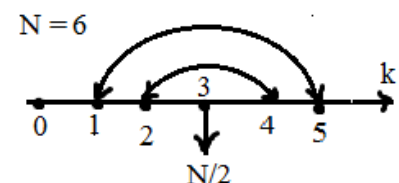
$$X(1) = 4 + 3W_6^1 + 2W_6^2 + W_6^3 = 3.5 - j4.33$$

$$X(2) = 4 + 3W_6^2 + 2W_6^{2(2)} + W_6^{3(2)} = 2.5 - j0.866$$

$$X(3) = 4 + 3W_6^3 + 2W_6^{2(3)} + W_6^{3(3)} = 2$$

$$X(4) = 4 + 3W_6^4 + 2W_6^{2(4)} + W_6^{3(4)} = 2.5 + j0.866 = X^*(2)$$

$$X(5) = 4 + 3W_6^5 + 2W_6^{2(5)} + W_6^{3(5)} = 3.5 + j4.33 = X^*(1)$$



Hence the given sequenced satisfies symmetry property.

(e) Circular Convolution – Multiplication of DFT's

If $x_1(n) \xleftarrow{\text{DFT}} X_1(k)$ and $x_2(n) \xleftarrow{\text{DFT}_N} X_2(k)$,

then $x_1(n) \otimes_N x_2(n) \xleftarrow{\text{DFT}_N} X_1(k).X_2(k)$.

Here $x_1(n) \otimes_N x_2(n)$ represents the circular convolution of $x_1(n)$ and $x_2(n)$.

Proof: By the definition of DFT

$$X_1(k) = \sum_{n=0}^{N-1} x_1(n) e^{-j\left(\frac{2\pi kn}{N}\right)} \quad \text{and} \quad X_2(k) = \sum_{p=0}^{N-1} x_2(p) e^{-j\left(\frac{2\pi kp}{N}\right)}$$

Let $X_3(k) = X_1(k) X_2(k)$

and $x_3(m)$ represents the sequence whose DFT is $X_3(k)$, then using IDFT

$$\begin{aligned} \text{DFT}^{-1}[X_3(k)] &= \frac{1}{N} \sum_{k=0}^{N-1} X_3(k) e^{j\left(\frac{2\pi km}{N}\right)} = \frac{1}{N} \sum_{k=0}^{N-1} X_1(k) X_2(k) e^{j\left(\frac{2\pi km}{N}\right)} \\ &= \frac{1}{N} \sum_{k=0}^{N-1} \left[\sum_{n=0}^{N-1} x_1(n) e^{-j\left(\frac{2\pi kn}{N}\right)} \sum_{p=0}^{N-1} x_2(p) e^{-j\left(\frac{2\pi kp}{N}\right)} \right] e^{j\left(\frac{2\pi km}{N}\right)} = \frac{1}{N} \sum_{n=0}^{N-1} x_1(n) \sum_{p=0}^{N-1} x_2(p) \sum_{k=0}^{N-1} e^{j\left(\frac{2\pi(m-n-p)}{N}\right)k} \end{aligned}$$

Using the identity $\sum_{k=0}^{N-1} a^k = \begin{cases} N & \text{for } a = 1 \\ \frac{1-a^N}{1-a} & \text{for } a \neq 1 \end{cases}$ here $a = e^{j\left(\frac{2\pi(m-n-p)}{N}\right)}$

$(m-n-p)$ is an integer multiple of N ; hence $a = e^{j\left(\frac{2\pi(m-n-p)}{N}\right)} = 1$ so that $\sum_{k=0}^{N-1} e^{j\left(\frac{2\pi(m-n-p)}{N}\right)k} = N$

Consider the second condition

$$\sum_{k=0}^{N-1} e^{j\left(\frac{2\pi(m-n-p)}{N}\right)k} = \frac{1 - e^{j\left(\frac{2\pi(m-n-p)}{N}\right)N}}{1 - e^{j\left(\frac{2\pi(m-n-p)}{N}\right)}} = \frac{1 - 1}{1 - e^{j\left(\frac{2\pi(m-n-p)}{N}\right)}} = 0 \quad \text{since } (m-n-p) \text{ is an integer}$$

Since $(m-n-p)$ is an integer and $e^{j2\pi(m-n-p)} = 1$ always

$$\text{DFT}^{-1}[X_3(k)] = \frac{1}{N} \left[\sum_{n=0}^{N-1} x_1(n) \sum_{p=0}^{N-1} x_2(p) \right] N = \left[\sum_{n=0}^{N-1} x_1(n) \sum_{p=0}^{N-1} x_2(p) \right] \quad \text{where } (m-n-p) \text{ is multiple of } N$$

Let $(m-n-p) = qN$ so that $p = (m-n-qN)$. Using this value in above equation

$$\sum_{p=0}^{N-1} x_2(p) = \sum_{n=0}^{N-1} x_2(m-n-qN) = \sum_{n=0}^{N-1} x_2(m-n, \text{mod } N) = \sum_{m=0}^{N-1} x_2((m-n))_N$$

$$\text{DFT}^{-1}[X_3(k)] = \sum_{n=0}^{N-1} x_1(n) \sum_{m=0}^{N-1} x_2((m-n))_N$$

$x_2(m - n + qN)$ represents a periodic sequence with period N delayed by n samples and is equivalent to sequence $x_2(m)$ shifted circularly by n samples and can be written as

$$x_2(m - n + qN) = x_2(m - n, \text{ modulo } N) = x_2(m - n)_N$$

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n) x_2(m - n)_N \quad \text{----- (A)}$$

Linear convolution between two sequences $x_1(n)$ and $x_2(n)$ can be defined as

$$y(n) = \sum_{k=-\infty}^{\infty} x_1(n) x_2(n - k) \quad \text{----- (B)}$$

Equation (A) is similar to equation (B) and in equation (A) sequence $x_2(n)$ is shifted circularly.

$$x_3(m) = x_1(n) \otimes_N x_2(n) = \sum_{n=0}^{N-1} x_1(n) x_2(m - n)_N \quad \text{where } m = 0, 1, 2, \dots, N - 1$$

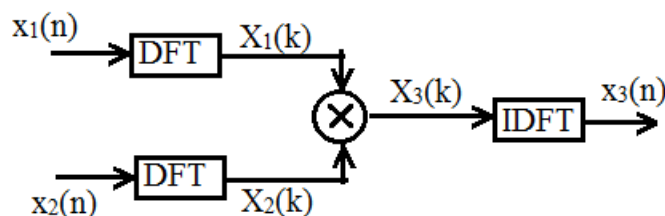
$$\text{Hence } x_1(n) \otimes_N x_2(n) \leftarrow \text{DFT}_N \rightarrow X_1(k).X_2(k)$$

Multiplication of two DFT's is equivalent to circular convolution of their sequence in time domain

Stockham's Method to calculate circular convolution

If $x_1(n) \leftarrow \text{DFT} \rightarrow X_1[k]$ and $x_2(n) \leftarrow \text{DFT} \rightarrow X_2[k]$, then $X_3[k] = X_1[k].X_2[k]$.

The circular convolution between $x_1(n)$ and $x_2(n)$ is given by $x_3(n) = \text{IDFT}\{X_3[k]\}$



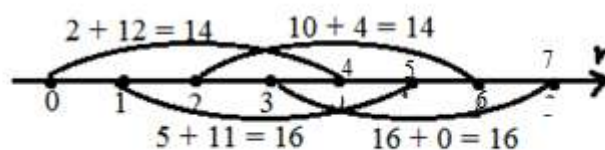
35 Find the circular convolution between the sequences $x_1(n) = \{ 2, 1, 2, 1 \}$ & $x_2(n) = \{ 1, 2, 3, 4 \}$

Solution: (a) Using linear convolution: $y(n) = x_1(n) \otimes x_2(n)$ Here $M = N = 4$

$x_1(n)$	2	1	2	1
$x_2(n)$	1	2	3	4
1	2	1	2	1
2	4	2	4	2
3	6	3	6	3
4	8	4	8	4

$$y(n) = \{ 2, 5, 10, 16, 12, 11, 4 \}$$

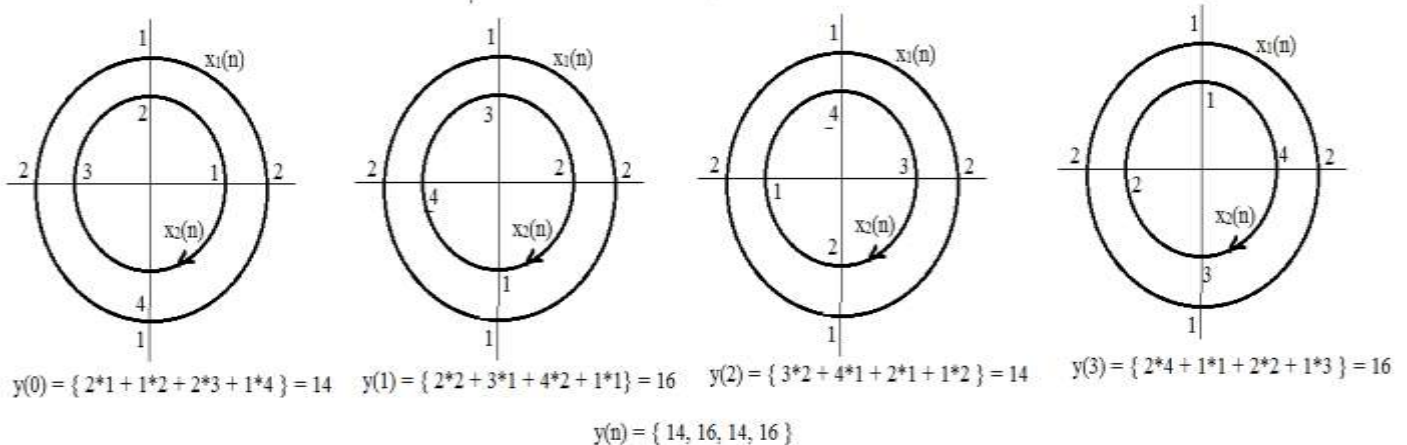
In circular convolution, the output has only $L = \max(M, N)$
Hence $L = \max(4, 4) = 4$ samples



$$y(n) = \{ 14, 16, 14, 16 \}$$

(ii) Using concentric circles method

keep $x_1(n)$ constant and move $x_2(n)$ one sample at a time in CW direction



(iii) Matrix method $y(n) = x_1(n) \otimes_4 x_2(n)$

$$y(n) = \begin{matrix} \downarrow \\ \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \end{bmatrix} \end{matrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2+2+6+4=14 \\ 1+4+3+8=16 \\ 2+2+6+4=14 \\ 1+4+3+8=16 \end{bmatrix}$$

$y(n) = \{14, 16, 14, 16\}$

(iv) Shifting method

$$y(n) = \sum_{m=0}^{N-1} x_1(m) x_2(m-n) = \sum_{n=0}^3 x_1(m) x_2(m-n) = \{14, 16, 14, 16\}$$

Index n	$x_1(m)$				$x_2(m-n)$				$y(n)$
0	2	1	2	1	1	4	3	2	$\{2*1 + 1*4 + 2*3 + 1*2\} = 14$
1	2	1	2	1	2	1	4	3	$\{2*2 + 1*1 + 2*4 + 1*3\} = 16$
2	2	1	2	1	3	2	1	4	$\{2*3 + 1*2 + 2*1 + 1*4\} = 14$
3	2	1	2	1	4	3	2	1	$\{2*4 + 1*3 + 2*2 + 1*1\} = 16$

36. Find the circular convolution between the sequences $x_1(n) = \{2, 1, 2, 1\}$ & $x_2(n) = \{1, 2, 3, 4\}$ using Stockham's method

Solution: Determine $X_1(k)$ and $X_2(k)$ using matrix method.

$$X_1(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 2 \\ 0 \end{bmatrix}$$

$X_1(k) = \{6, 0, 2, 0\}$

$$X_2(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 \\ -2+j2 \\ -2 \\ -2-j2 \end{bmatrix}$$

$X_2(k) = \{10, (-2+j2), -2, (-2-j2)\}$

$$Y(k) = X_1(k) X_2(k) = \{ X_1(0) X_2(0), X_1(1) X_2(1), X_1(2) X_2(2), X_1(3) X_2(3) \} = \{ 60, 0, -4, 0 \}$$

Taking IDFT on both sides

$$y(n) = \frac{1}{N} \sum_{k=0}^{N-1} Y(k) W_N^{-kn} = \frac{1}{4} \sum_{k=0}^3 Y(k) W_4^{-kn} = \{14, 16, 14, 16\}$$

$$y(n) = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 60 \\ 0 \\ -4 \\ 0 \end{bmatrix} = \begin{bmatrix} 14 \\ 16 \\ 14 \\ 16 \end{bmatrix}$$

$y(n) = \{ 14, 16, 14, 16 \}$

37. Compute circular convolution using DFT/IDFT for $x_1(n) = n$ for $0 \leq n \leq 3$ and $x_2(n) = \cos(\pi n/2)$

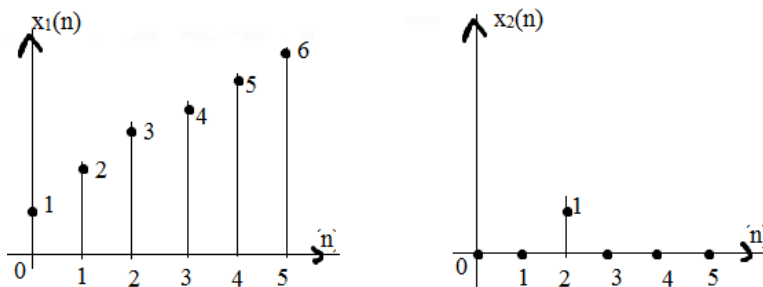
Solution: $N = 4$ $x_1(n) = \{ 0, 1, 2, 3 \}$ $x_2(n) = \{ 1, 0, -1, 0 \}$

Find $X_1(k)$ and $X_2(k)$. Take the product $Y(k) = X_1(k) \cdot X_2(k)$

$$X_1(k) = \{ 6, -2 + j2, -2, -2 - j2 \} \quad X_2(k) = \{ 0, 2, 0, 2 \}$$

$$y(n) = \text{IDFT} [Y(k)] = \{ -2, -2, 2, 2 \} \quad \text{Refer problem 36}$$

38. Find the circular convolution between the sequences shown below



Solution: $x_1(n) = \{ 1, 2, 3, 4, 5, 6 \}$ $x_2(n) = \{ 0, 0, 1, 0, 0, 0 \} = \delta(n - 2)$

$$y(n) = x_1(n) \otimes_N x_2(n) = x_1(n) \otimes_N \delta(n - 2) = x_1((n - 2))_N = x_1((n - 2))_6$$

Shift $x(n)$ by two samples in ACW direction $y(n) = \{ 5, 6, 1, 2, 3, 4 \}$

39. Given $x(n) = \{ 1, 2, 3, 4 \}$; $h(n) = \{ 1, 2, 2 \}$ Compute (i) Circular convolution (ii) Linear convolution (iii) Linear convolution from circular convolution. Comment on the result

Solution: (i) Circular Convolution: $x(n) = \{ 1, 2, 3, 4 \}$ $N = 4$; $h(n) = \{ 1, 2, 2 \}$ $M = 3$

Make the length of the sequences $L = \max(M, N) = \max(3, 4) = 4$

Hence $h(n) = \{ 1, 2, 2, 0 \}$ Let $y(n) = x(n) \otimes_4 h(n)$

$$y(n) = \begin{bmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 15 \\ 12 \\ 9 \\ 14 \end{bmatrix}$$

$$y(n) = \{ 15, 12, 9, 4 \}$$

(ii) Linear convolution $y(n) = x(n) \otimes h(n)$

$$y(n) = \{ 1, 4, 9, 14, 14, 8 \}$$

(iii) Linear convolution using Circular convolution

Modify the length of the sequences to $L = M + N - 1 = 4 + 3 - 1 = 6$

$$x(n) = \{ 1, 2, 3, 4, 0, 0 \} \quad h(n) = \{ 1, 2, 2, 0, 0, 0 \} \quad y(n) = x(n) \otimes_6 h(n) = \{ 1, 4, 9, 14, 14, 8 \}$$

$$y(n) = \begin{bmatrix} 1 & 0 & 0 & 4 & 3 & 2 \\ 2 & 1 & 0 & 0 & 4 & 3 \\ 3 & 2 & 1 & 0 & 0 & 4 \\ 4 & 3 & 2 & 1 & 0 & 0 \\ 0 & 4 & 3 & 2 & 1 & 0 \\ 0 & 0 & 4 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 9 \\ 14 \\ 14 \\ 8 \end{bmatrix}$$

$$y(n) = \{ 1, 4, 9, 14, 14, 8 \}$$

Comment: The outputs Circular convolution and Linear convolution are same if the sequence lengths of input sequences of circular convolution are $L = M + N - 1$

40. Compute the circular convolution using DFT + IDFT for the following sequences $x_1(n) = \{2, 3, 1, 1\}$ and $x_2(n) = \{1, 3, 5, 3\}$

Solution: Similar to problem 36

41. Find the circular convolution between the sequences

$$x_1(n) = e^{j\pi n} ; 0 \leq n \leq 7 \text{ and } x_2(n) = u(n) - u(n - 5). \text{ Use time domain approach}$$

Solution: $x_1(n) = e^{j\pi n} = \{ 1, -1, 1, -1, 1, -1, 1, -1 \} \quad M = 8$

$$x_2(n) = u(n) - u(n - 5) = \{ 1, 1, 1, 1, 1 \} \quad N = 5$$

For circular convolution make sequence lengths $L = \max(M, N) = \max(8, 5) = 8$

$$x_2(n) = \{ 1, 1, 1, 1, 1, 0, 0, 0 \}$$

$$y(n) = x_1(n) \otimes_8 x_2(n)$$

$$y(n) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$y(n) = \{1, -1, 1, -1, 1, -1, 1, -1\}$$

42 Determine the N point DFT of $x(n) = \cos\left(\frac{2\pi n}{N}\right)$ and hence find $x(n) \otimes_N x(n)$

Solution: $x(n) = \cos\left(\frac{2\pi n}{N}\right) = \frac{e^{j2\pi n/N} + e^{-j2\pi n/N}}{2} = \left[\frac{W_n^{-n} + W_n^n}{2}\right]$

$$X(k) = \text{DFT} \left[\frac{W_n^{-n} + W_n^n}{2} \right] = N \left[\frac{\delta(k+1) + \delta(k-1)}{2} \right]$$

$$X(k) \cdot X(k) = N^2 \left[\frac{\delta(k+1) + \delta(k-1)}{4} \right]^2 = \frac{N^2}{2} \left[\frac{\delta(k+1) + \delta(k-1)}{2} \right]$$

Let $y(n) = x(n) \otimes_N x(n)$ hence $y(n) = \text{IDFT} [X(k) \cdot X(k)]$

$$y(n) = \text{IDFT} \left\{ \frac{N^2}{2} \left[\frac{\delta(k+1) + \delta(k-1)}{2} \right] \right\} = \frac{N}{2} \cos\left(\frac{2\pi n}{N}\right)$$

43 Find the circular convolution between the sequences $x_1(n) = \cos\left(\frac{2\pi n}{N}\right)$ and $x_2(n) = \sin\left(\frac{2\pi n}{N}\right)$

Solution: $x_1(n) = \cos\left(\frac{2\pi n}{N}\right) = \frac{e^{j2\pi n/N} + e^{-j2\pi n/N}}{2} = \left[\frac{W_n^{-n} + W_n^n}{2}\right]$

$$X_1(k) = \text{DFT} \left[\frac{W_n^{-n} + W_n^n}{2} \right] = N \left[\frac{\delta(k-1) + \delta(k+1)}{2} \right]$$

$$x_2(n) = \sin\left(\frac{2\pi n}{N}\right) = \frac{e^{j2\pi n/N} - e^{-j2\pi n/N}}{2j} = \left[\frac{W_n^{-n} - W_n^n}{2j}\right]$$

$$X_2(k) = \text{DFT} \left[\frac{W_n^{-n} - W_n^n}{2j} \right] = N \left[\frac{\delta(k-1) - \delta(k+1)}{2j} \right]$$

$$X_1(k) \cdot X_2(k) = N^2 \left(\frac{[\delta(k-1)]^2 - [\delta(k+1)]^2}{4j^2} \right) = \left(\frac{N}{2}\right) N \left[\frac{\delta(k-1) - \delta(k+1)}{2j} \right]$$

Let $y(n) = x_1(n) \otimes_N x_2(n)$ hence $y(n) = \text{IDFT} [X_1(k) \cdot X_2(k)]$

$$y(n) = \left(\frac{N}{2}\right) \text{IDFT} \left\{ N \left[\frac{\delta(k-1) - \delta(k+1)}{2j} \right] \right\} = \frac{N}{2} \sin\left(\frac{2\pi n}{N}\right)$$